

torch.squeeze#

<https://pytorch.org/docs/stable/generated/torch.squeeze.html>

Description:

Returns a tensor with all specified dimensions of input of size 1 removed.

For example, if input is of shape: $(A \times 1 \times B \times C \times 1 \times D)$ then the `input.squeeze()` will be of shape: $(A \times B \times C \times D)$. When `dim` is given, a squeeze operation is done only in the given dimension(s).

If input is of shape: $(A \times 1 \times B)$, `squeeze(input, 0)` leaves the tensor unchanged, but `squeeze(input, 1)` will squeeze the tensor to the shape $(A \times B)$. The returned tensor shares the storage with the input tensor, so changing the contents of one will change the contents of the other.

Function Signature

```
torch.squeeze(input: Tensor, dim: Optional[Union[int, List[int]]]) → Tensor#
```

Parameters

if given, the input will be squeezed

Code Examples

```
>>> x = torch.zeros(2, 1, 2, 1, 2)
>>> x.size()
torch.Size([2, 1, 2, 1, 2])
>>> y = torch.squeeze(x)
>>> y.size()
torch.Size([2, 2, 2])
>>> y = torch.squeeze(x, 0)
>>> y.size()
torch.Size([2, 1, 2, 1, 2])
>>> y = torch.squeeze(x, 1)
>>> y.size()
torch.Size([2, 2, 1, 2])
>>> y = torch.squeeze(x, (1, 2, 3))
torch.Size([2, 2, 2])
```

Notes & Warnings

NOTE

Note The returned tensor shares the storage with the input tensor, so changing the contents of one will change the contents of the other.

⚠ WARNING

Warning If the tensor has a batch dimension of size 1, then `squeeze(input)` will also remove the batch dimension, which can lead to unexpected errors. Consider specifying only the dims you wish to be squeezed.

Detailed Documentation

Documentation

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For example, if input is of shape: $(A \times 1 \times B \times C \times 1 \times D)$ then the `input.squeeze()` will be of shape: $(A \times B \times C \times D)$.

When `dim` is given, a squeeze operation is done only in the given dimension(s). If input is of shape: $(A \times 1 \times B)$, `squeeze(input, 0)` leaves the tensor unchanged, but `squeeze(input, 1)` will squeeze the tensor to the shape $(A \times B)$.

The returned tensor shares the storage with the input tensor, so changing the contents of one will change the contents of the other.

If the tensor has a batch dimension of size 1, then `squeeze(input)` will also remove the batch dimension, which can lead to unexpected errors. Consider specifying only the dims you wish to be squeezed.

`input (Tensor)` – the input tensor.

`dim (int or tuple of ints, optional)` –

only in the specified dimensions.

Changed in version 2.0: `dim` now accepts tuples of dimensions.

